Magnitude, Not Shape: Volatility Beats Downside- and Tail-Risk Metrics at Forecasting Drawdown

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ABSTRACT

Every few years a new downside- or tail-risk metric promises to improve on plain return volatility. The premise is always the same. The *shape* of the loss distribution, whether its asymmetry, tail heaviness, or co-crash structure, is supposed to carry information that symmetric volatility misses. We test that premise as adversarially as the data allow. On a survivorship-free, multi-regime, multiple-testing-corrected evaluation of ten metrics across five test beds, no metric reliably beats volatility at forecasting an asset's forward maximum drawdown, whether the metric is published or machine-discovered. The failure has a clean anatomy, which we call *magnitude, not shape*. The partial rank correlation of Value-at-Risk with forward drawdown, controlling for volatility, survives Benjamini–Hochberg correction on all five beds, though it reaches only 0.19 at its largest. Every tail-shape metric adds nothing or subtracts. Yet the shape metrics are not broken. On genuinely shape-sensitive targets they beat volatility. Drawdown is simply a magnitude phenomenon that shape does not drive. Volatility is hard to beat because it is persistent, but a predictive mediation test indicates it is not a sufficient statistic. The tiny magnitude edge is not economically free either. A cost-aware backtest and a pre-registered composite add nothing over volatility except in fat-tailed crypto. Finally, an autonomous large-language-model discovery loop only inflates validation while its locked-test score plateaus at volatility, a data-snooping caution now that agentic search makes signal-hunting nearly free. Volatility is not glamorous, but on a level field it is very hard to beat.

CCS CONCEPTS

• **Computing methodologies → Machine learning**; • **Applied computing → Economics**.

KEYWORDS

downside risk, tail risk, drawdown, volatility, survivorship bias, multiple testing, data snooping, LLM agents

1 INTRODUCTION

Prices fall faster than they rise, and it is the fall that ruins investors. From that intuition a large literature has grown. It proposes metrics that look past symmetric volatility to the shape of the loss distribution. These include lower partial moments and the Sortino ratio [6, 15, 22], the partial-moment "predictive" ratio (UPM/LPM) [23], left-tail Value-at-Risk and expected shortfall [3, 21], realized semibetas [8], downside beta [2, 18], lower-tail dependence [9], disappointment-aversion risk (GDA) [13], and tail-exponent factors [17]. Each rests on the same premise, that asymmetry or tail structure carries information volatility misses.

We test that premise on the fairest field we can build. Three disciplines the source literature usually omits turn out to matter enormously. A *survivorship-free* universe keeps the assets that actually died. A *locked* out-of-sample holdout stops search from leaking into the verdict. And *multiple-testing correction* keeps an evaluation of dozens of comparisons from manufacturing winners. We evaluate ten metrics, volatility and nine downside- and tail-risk challengers, as forecasters of 90-day forward maximum drawdown across five cross-sectional test beds spanning calm, bull, and crash regimes in crypto and equity, with per-date rank correlations, paired block-bootstrap confidence intervals, and Benjamini–Hochberg (BH) correction.

The headline is negative and, in hindsight, unsurprising. No metric reliably beats plain volatility. Our contribution is the anatomy of that result and the integrity of the evaluation behind it. (1) A clean empirical law we call magnitude, not shape. The only signal surviving correction beyond volatility is the *size* of tail losses (VaR/ES/downside deviation), not their asymmetry (§4). (2) Volatility is hard to beat because it is persistent, though a predictive test shows forward volatility is not a sufficient statistic (§5). (3) The residual edge is not economically free, except in fat-tailed crypto (§5). (4) An autonomous large-language-model (LLM) discovery loop does not beat volatility out-of-sample and only overfits validation, a data-snooping caution as agentic search makes signal-hunting nearly free (§5). Every experiment is retained, and supporting detail is in the appendices.

Related work. Lower partial moments generalize semivariance [6, 15] and underpin the Sortino ratio [22] and the partial-moment ratio [23]; tail measures include VaR/ES [3, 21], semibetas [8], downside beta [2, 18], lower-tail dependence [9], disappointment aversion [13], and tail exponents [17]. Most were built to price the cross-section, not forecast one asset's drawdown; a separate literature studies the drawdown statistic directly [10, 19]. Volatility itself is a persistent, tradable risk signal [14, 20]. That volatility is persistent is the founding fact of ARCH/GARCH [1, 7, 11, 12]; that searching many signals inflates in-sample rank correlations is classical data snooping [4, 5, 16, 24]. Our novelty is to show an *autonomous* agent reaches the snooping frontier at near-zero cost.

2 EXPERIMENTAL SETUP

Five test beds. *Crypto (survivorship-free):* a CoinMarketCap daily panel (2013–2018) that retains crashed and delisted coins, so a point-in-time top-100-by-market-cap universe includes assets that later crater; we use three regimes: 2016 (calm), 2017 (bull), 2018 (crash). *Equity:* S&P 500 constituents, 1994–1999 (calm) and 2005–2009 (GFC). The equity beds are survivorship-biased by construction (freely available equity data retains almost no true deaths, itself a telling finding), so the survivorship claim rests on the crypto beds, while equity tests cross-asset generality. We checked this directly. A public, delisting-reputed price dump (the

Kaggle Huge Stock Market dataset) contains none of our windows' famous S&P 500 casualties, with Lehman, Bear Stearns, Enron, and WorldCom all absent, and yields zero mid-window delistings, so no free-data equity bed is genuinely survivorship-free.

What we forecast. From a 180-day trailing window we forecast each asset's 90-day forward maximum drawdown, coding a delisting as a −100% loss. We score a metric by its per-date cross-sectional rank correlation with that drawdown, averaged over the bed, cross-sectional rather than time-series because the operational question is which assets to derisk on a given date.

3 THE VERDICT: NOTHING BEATS VOLATILITY

The verdict is blunt. No metric beats volatility in a way that survives a change of asset class. Table 1 ranks each metric against volatility standalone, with significance from paired block-bootstrap 95% intervals and Benjamini–Hochberg (BH) correction at $q = 0.05$ applied within each family (head-to-head and partials, ~45 cells each). The block bootstrap uses length-3 blocks over the per-date series. On the thin crypto beds (7–9 dates) this grid is coarse, so we back every crypto star with its interval and minimum detectable effect (App. B) and read crypto-only significance as suggestive. The only metrics that significantly beat volatility on any bed are two volatility variants, downside deviation and VaR, on the single crypto-bull bed, and both reverse on equity, where volatility significantly beats them. Every genuinely different, shape-based metric is significantly worse than volatility on every bed.

We fix the bar in advance to avoid a moving goalpost. An edge counts as *generalizable* only if a metric BH-significantly beats volatility on at least one crypto and one equity bed, i.e., survives a change of asset class. No metric clears it. Small, regime-specific edges exist for volatility's own variants, but none generalizes. We pre-registered the main tests, fixing them before we saw any results so we cannot cherry-pick them, namely the verdict, the magnitude-versus-shape increments, and the composite. The finer slices by market stress and across beds were explored afterward and count only as suggestive.

4 THE ANATOMY: MAGNITUDE, NOT SHAPE

If nothing beats volatility, what is left to forecast beyond it? The *partial* rank correlation (Spearman $\rho(\text{metric}, \text{drawdown} \mid \text{volatility})$, which isolates what a metric adds beyond volatility) answers this. Figure 1 plots it for every metric and bed, and the picture splits cleanly by *what a metric measures*. Tail-magnitude measures carry a small but genuine increment (VaR significant on **all five beds**, partial ρ up to 0.19; downside deviation on the three crypto beds), while tail-shape measures (the Viole–Nawrocki ratio, lower-tail dependence, the tail exponent, disappointment aversion) are ≈ 0 or negative everywhere. This is not merely an *own-asset vs systematic* artifact. The one own-asset shape metric (the Hill tail exponent) is itself ≈ 0/negative, so magnitude beats shape within the own-asset family, while no systematic co-movement metric adds signal (Figure 1 groups both axes). The increment is also not an artifact of the −100% delisting code. Mid-window delistings are ≤1% of asset-dates, and valuing them at last price or dropping them moves VaR's increment by ≤0.02 (App. G). Table 2 gives the exact increments with BH significance; power checks are in App. B.

VaR is worse than volatility standalone on equity (Table 1) yet has a positive increment there. As a standalone ranker it is a noisier proxy for volatility, but the partial correlation isolates its small orthogonal component. The increment is also larger where markets are fat-tailed and blow-up-prone (crypto increments 0.12–0.19, excess kurtosis 8–21, vs equity 0.03–0.08; App. F). Tail magnitude carries incremental information precisely when the tail is active.

One might object that drawdown is a magnitude-dominated statistic, so a magnitude metric is bound to win. Expected maximum drawdown scales with $\sigma\sqrt{T}$ [19], so the objection is correct, and it is the point. Drawdown is the loss that ruins investors, and the literature proposes shape metrics to forecast it. When we forecast genuinely shape-sensitive targets instead, forward skew, downside fraction, and tail-exceedance frequency, the shape metrics do beat volatility on nearly every bed, while volatility predicts them poorly (App. J). Shape metrics forecast shape. Shape is simply not what drives drawdown.

Table 1: Standalone out-of-sample drawdown-forecast Spearman correlation. $*$ = significantly beats volatility, † = significantly worse (both survive BH, $q = 0.05$); unmarked = indistinguishable. The Hill tail exponent [17] is omitted here (not estimable per date on equity); its increment appears in Table 2. Bold marks the highest value in each column.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
volatility (base)	**0.62**	0.36	0.38	**0.61**	**0.56**
downside deviation	**0.62**	0.38$*$	**0.41**	0.58†	0.54†
VaR	0.61	**0.40**$*$	0.36	0.58†	0.54†
ES [21]	**0.62**	0.35	0.33†	0.58†	0.52†
neg. semibeta [8]	0.40†	0.26†	0.31†	0.49†	0.46†
downside beta [2]	0.01†	0.09†	0.15†	0.24†	0.34†
lower-tail dep. [9]	−0.38†	−0.15†	−0.29†	−0.07†	−0.05†
GDA [13]	−0.06†	−0.11†	−0.05†	−0.24†	−0.32†
UPM/LPM [23]	−0.55†	−0.28†	−0.35†	−0.43†	−0.42†

Table 2: Partial Spearman $\rho(\text{metric}, \text{drawdown} \mid \text{volatility})$. Rows 1–3 tail magnitude, rows 4–9 tail shape. $*$ survives BH ($q = 0.05$, positive); † survives BH negative; n/e = not estimable per date. Bold marks the highest value in each column.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
VaR	+0.12$*$	**+0.19**$*$	+0.13$*$	**+0.03**$*$	**+0.08**$*$
downside dev.	+0.13$*$	+0.14$*$	**+0.20**$*$	+0.02	+0.01
ES [21]	**+0.14**$*$	+0.08$*$	+0.08	**+0.03**	+0.00
neg. semibeta [8]	+0.02	+0.08$*$	+0.13$*$	**+0.03**	+0.05$*$
UPM/LPM [23]	−0.00	+0.01	−0.05	−0.05†	−0.07†
downside beta [2]	−0.08†	+0.03	+0.02	+0.01	+0.04
Hill tail [17]	−0.09†	−0.13†	−0.09†	n/e	n/e
lower-tail dep. [9]	−0.07	−0.04	−0.17†	−0.01	+0.00
GDA [13]	+0.03	−0.05	+0.05	−0.01	−0.03

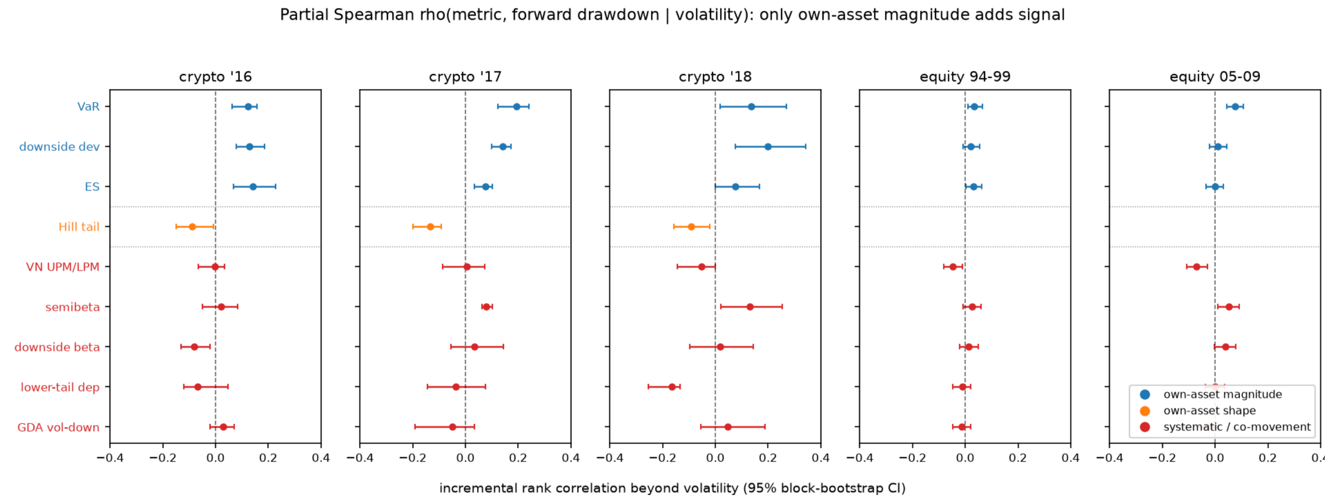

Figure 1: Incremental signal beyond volatility, by bed: partial Spearman $\rho(\text{metric}, \text{drawdown} \mid \text{volatility})$, grouped on both axes a skeptic cares about, namely what the metric measures (magnitude vs shape) and whose distribution it uses (own-asset vs systematic co-movement with the market). Only own-asset magnitude (blue) sits reliably right of zero. The lone own-asset shape metric, the Hill tail exponent (orange), is ≈ 0 or negative, so "magnitude beats shape" holds within the own-asset family, and no systematic co-movement metric (red) adds signal. Whiskers are 95% block-bootstrap CIs; dotted lines separate the three families.

Table 3: Persistence chain (mean per-date Spearman; all CIs exclude 0).

Bed	trail→fwd vol	fwd vol→dd	(trail→dd)
crypto-16	0.68	0.77	0.62
crypto-17	0.42	0.62	0.36
crypto-18	0.42	0.53	0.38
equity 94	0.84	0.71	0.61
equity 05	0.80	0.70	0.56

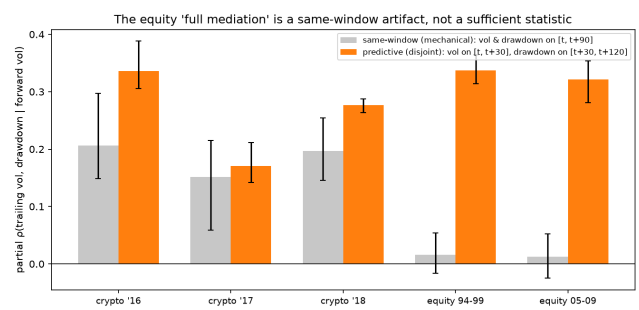

Figure 2: Mediation: partial $\rho(\text{trailing vol}, \text{drawdown} \mid \text{forward vol})$. The same-window (mechanical) design gives ≈ 0 on equity, falsely suggesting full mediation. The predictive (disjoint-window) design gives 0.32–0.34, so forward volatility does not fully screen off trailing volatility.

5 WHY VOLATILITY WINS, AND CAN ANYTHING BEAT IT?

5.1 Persistence

Volatility wins because it is persistent, the founding fact of ARCH/GARCH [7, 12]. Trailing volatility forecasts forward volatility (rank correlation 0.42–0.84), which in turn drives forward drawdown. Table 3 decomposes this chain by bed. Both links exceed the direct trailing-vol→drawdown rank correlation they compose. We resist calling forward volatility a *sufficient statistic*. A naive same-window mediation suggests it, but that test is contaminated (forward volatility and drawdown share a window). Measured on a disjoint, predictive window, trailing volatility retains a partial correlation of 0.32–0.34 on equity (Figure 2), so persistence is a leading but incomplete explanation, not a mechanism that reduces drawdown to one number.

5.2 Does the edge pay?

A rank correlation is not a profit-and-loss statement. The magnitude-not-shape split of §4, volatility plus a single tail-magnitude term, implies one natural composite, the equal-weight sum $z(\text{vol}) + z(\text{VaR})$ with no fitting, where $z(\cdot)$ is the cross-sectional z-score (each metric standardized across the assets on a given date). We pre-registered it. On a locked panel it shows no significant out-of-sample gain over volatility (Newey–West $t = 0.67$, lag 3), beating it only on fat-tailed crypto-bull (Table 4). A cost-aware tercile backtest agrees. The high-risk tercile realizes 10–30 percentage points more forward drawdown than the low-risk tercile on every bed, net of a 20 bps turnover cost, and the composite adds nothing net of turnover except, again, on

Table 4: Composite $z(\text{vol}) + z(\text{VaR})$ vs. volatility: per-date Spearman and difference (95% CI); significant only on crypto-bull.

Test bed	vol ρ	comp ρ	difference (95% CI)
crypto-16	0.62	0.62	-0.00 (n.s.)
crypto-17	0.36	0.40	$+0.05$ $[0.02, 0.07]$
crypto-18	0.38	0.39	$+0.01$ (n.s.)
equity 94	0.61	0.60	-0.008 $[-0.011, -0.004]$
equity 05	0.56	0.56	-0.00 (n.s.)
Locked panel: 0.376 $\rightarrow$ 0.388, Newey–West $t = 0.67$ (n.s.).			

crypto-bull. Volatility is the workhorse. Magnitude adds a little, and only where tails are fat.

5.3 Can a machine find more?

If the residual is this small, a tireless automated search should fail out-of-sample and merely overfit validation. We test this on a strict train/validation/locked-test split, the locked test scored once after the metric is frozen. An LLM agent rewriting a scoring function and a 20,000-configuration sweep both inflate the best *validation* score with search intensity while the *locked-test* score plateaus at the volatility baseline (Figure 3). Across 12 holdout panels $\times$ 3 search seeds (App. K, a non-LLM generator at scale, with the LLM loop as the single original panel), the validation-selected metric's out-of-sample edge over volatility is at most +0.07 and averages +0.02–0.03, straddling zero. No search, LLM or brute-force, finds a genuine improvement. A non-LLM random sweep reproduces the plateau, so the danger is the cheap search, not the language model. The statistics are classical data snooping [5, 24]; what is new is that an autonomous agent removes the last brake on trial count (analyst effort), so a locked, survivorship-free holdout scored exactly once becomes mandatory, not merely advisable.

5.4 What is volatility for?

The clearest positive use is forecasting outright blow-ups. On the locked crypto test, volatility predicts which assets crater (forward return $\leq -80\%$) with ROC-AUC 0.68 (0.95 for the stricter $\leq -90\%$ cutoff) and a 2.5× top-decile lift, while a gradient-boosting model over all ten metrics, chosen on validation (AUC 0.73), overfits and loses on the locked test (AUC 0.59). The blow-up base rate is only $\approx$1.2%, so these AUCs are noisy and we read them as ordinal. Again the size, not the shape, of recent moves carries the signal. This is a ranking result, not a deployment recommendation (cost, liquidity, and reflexivity frictions are unmodeled).

6 DISCUSSION AND CONCLUSION

The pieces form one coherent picture. Volatility sets a high bar because it is persistent (§5). The only thing left to add is the magnitude of tail losses, and only a little, mostly in fat-tailed markets (§4). So no standalone metric beats volatility (§3), a pre-registered composite yields no significant out-of-sample gain, automated search cannot find one, and the economics track the statistics (§5). The recurring failure of asymmetry and shape metrics (including the partial-moment ratio [23]) is not an implementation accident. Once symmetric volatility is accounted for, the asymmetric structure adds no incremental cross-sectional information for forecasting drawdown. The fair-field disciplines change what one would report, not the verdict. Survivorship bias alone flips a co-crash shape metric from significantly worse than volatility to a tie and inflates the apparent magnitude edge (App. I), yet the verdict that no shape metric beats volatility is robust to dropping any single discipline. The disciplines guard the honesty of the numbers, not the direction of the conclusion.

We are deliberately careful about the strength of the persistence story. An earlier version of this project claimed forward volatility was a sufficient statistic, found that claim was a same-window measurement artifact, and retracted it here, just as an earlier "survivorship masks downside value" thesis was retracted when a hardened test refuted it. The paper reports only claims that survived stress-testing.

The practical takeaway is plain but firm. For cross-sectional drawdown forecasting, use volatility (optionally with a small tail-magnitude term in fat-tailed markets) and treat any downside metric "discovered" to beat it, by hand or by machine, as unproven until it clears a locked, survivorship-free holdout. On a level field, volatility is not glamorous, but it is very hard to beat.

Limitations. The performance measure is rank correlation, not a full tradable P&L (§5 is a first economic check, not a portfolio study); the crypto beds have few formation dates (7–9), so their significance is suggestive and the null "shape adds nothing" leans on the well-powered equity beds; and the repurposed pricing metrics are daily adaptations, so results speak to drawdown forecasting, not the originals' pricing claims.

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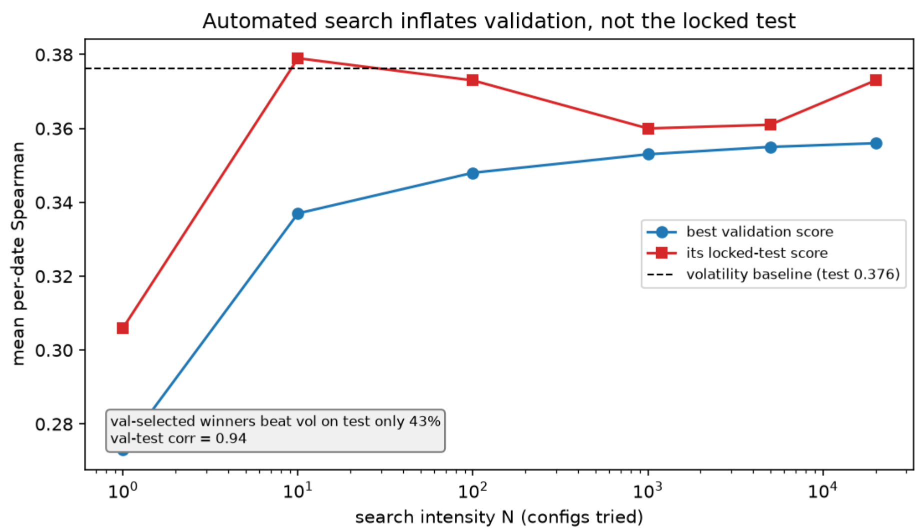

Figure 3: Automated discovery does not beat volatility out of sample. As search intensity grows, the best validation score climbs while its locked-test score stays flat at the volatility baseline. On this ($2017 \rightarrow 2018$-crash) holdout, validation-selected winners beat volatility on the locked test only 43% of the time; across 12 panels $\times$ 3 seeds the win-rate is regime-dependent, but the locked-test edge over volatility never exceeds noise (App. K).

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A REPRODUCIBILITY

Code, data pointers, and result logs to reproduce every table and figure are available at https://github.com/meghanadhpulivarthi/downside-risk-metrics-study. The data sources are a CoinMarketCap daily crypto dump (retaining delisted coins), yfinance S&P 500 constituents and the market index, and point-in-time index membership.

B POWER AND PLACEBO

The partial correlations of Table 2 (§4) survive BH on the beds marked; the one equity ES cell that is nominally significant does not survive correction. We also report power on the thin crypto beds. For VaR, on crypto-16 ($n = 8$) and crypto-17 ($n = 9$) the effect (0.12, 0.19) exceeds the 80%-power minimum detectable effect (0.09, 0.09) and is stable under leave-one-date-out; on crypto-18 ($n = 7$) the effect (0.13) is below its MDE (0.18) and is reported as suggestive. A placebo (permuting forward drawdown within each date) drives the trailing-volatility correlation to ≈ 0 on four of five beds; the exception, crypto-18 (+0.028, CI [0.003, 0.059]), is negligible against a real rank correlation of ≈ 0.38 and reflects the coarse block bootstrap at $n = 7$. On the well-powered equity beds the 80%-power minimum detectable increment for the shape metrics is 0.034–0.042, and the observed shape increments sit at or below it (Table 2), so on equity "shape adds nothing" is a powered null rather than a failure to detect.

C THE PERSISTENCE CHAIN

Table 3 (§5) decomposes why trailing volatility forecasts drawdown. The forward-vol→drawdown column is contemporaneous (both over the same forward window) and hence partly mechanical [1]; the genuinely predictive link is trailing→forward volatility. Figure 4 shows the two links relative to the direct trailing-volatility→drawdown rank correlation, by bed.

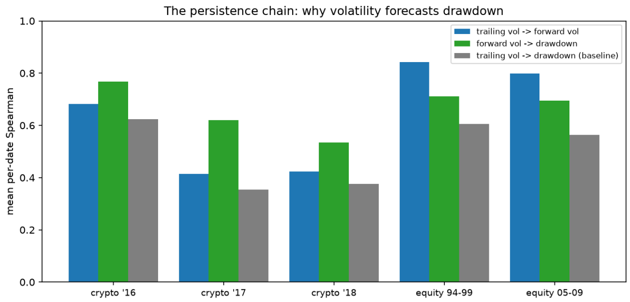

Figure 4: The two links of the persistence chain relative to the direct trailing-volatility→drawdown rank correlation, by bed.

D MEDIATION: FORWARD VOLATILITY IS NOT A SUFFICIENT STATISTIC

It is tempting to claim drawdown predictability simply *is* volatility persistence, meaning that forward volatility is a sufficient statistic. A naive mediation test seems to confirm it. Controlling for forward volatility, trailing volatility's partial correlation with drawdown collapses to ≈ 0 on equity. But that test is contaminated. Forward volatility and forward drawdown are measured over the *same* window, so conditioning on one mechanically absorbs the other. Measuring the mediator on a *disjoint, earlier* window (volatility over days 1–30, drawdown over days 31–120, so it is genuinely predictive), the mediation vanishes. Trailing volatility retains a partial correlation of 0.32–0.34 on equity (Figure 2). Trailing volatility carries forecasting content a short-horizon volatility forecast does not, so persistence is a leading but incomplete explanation. Because the disjoint mediator is a noisy 30-day volatility proxy, measurement error could inflate this residual, so we read it as consistent with incomplete mediation rather than proof that forward volatility is insufficient. Consistently, drawdown rank correlation and volatility persistence rise and fall together as the horizon varies (App. E). The mediation figure is Figure 2 in §5.

E HORIZON DYNAMICS

Sweeping the forecast horizon from 14 to 90 days, volatility persistence and drawdown rank correlation move together (Figure 5). On equity both rise (persistence $0.66 \rightarrow 0.84$, rank correlation $0.47 \rightarrow 0.61$, at a roughly constant gap), while on crypto both are flat. Wherever persistence is higher, so is the rank correlation, consistent with persistence driving most of the forecastable variation, with the constant gap the part it does not explain.

F WHEN THE MAGNITUDE INCREMENT IS LARGEST

Is the VaR increment genuinely larger where tails are fatter, or does it only look that way across five points? We regress each formation date's VaR increment, the partial $\rho(\mathrm{VaR}, \mathrm{drawdown} \mid \mathrm{volatility})$, on that date's cross-sectional tail-fatness, pooling all 144 dates across the five beds. Because the relationship is between beds, the honest test clusters by bed, where the slope is positive but only suggestive at five clusters (median excess kurtosis $p \approx 0.10$, blow-up rate $p \approx 0.15$). Treating the 144 dates as independent, which they are not, the same slopes (+0.013 for kurtosis, +0.76 for blow-up rate) exclude zero, so the within-regime relationship is consistent even where the between-bed test is underpowered. Figure 6 shows the per-bed picture, with the increment concentrated in the fat-tailed crypto beds.

The same pattern holds within crypto. Splitting the crypto dates at their median aggregate (cross-sectional-median) volatility, the VaR increment rises from +0.13 (CI $[0.06, 0.18]$) on low-stress dates to +0.18 ($[0.09, 0.24]$) on high-stress dates (24 dates, 12 per regime; suggestive given the overlap), while on equity it is flat and small (+0.056 vs. +0.050; 120 dates). Within crypto as across beds, then, the increment grows precisely when the tail turns active.

G DELISTING-CODING ROBUSTNESS

The crypto forward-drawdown codes a mid-window delisting as a terminal −100% loss. To confirm the magnitude increment is a genuine crash-severity signal, not an artifact of that code, nor a restatement of the blow-up result (§5), we recompute $\rho(\mathrm{metric}, \mathrm{drawdown} \mid \mathrm{vol})$ under two alternative codings. The first values a delisted asset at its *last traded price* (ordinary peak-to-trough, no −100% override). The second *excludes* delisted assets (rank correlation among survivors only). Within the point-in-time top-100 universe, mid-window delistings are rare (0.9% of asset-dates on crypto-2016 and none on the other four beds), so the three codings barely differ. VaR's increment is unchanged to within ±0.02 across all three (Table 5); downside deviation and ES behave identically. The magnitude edge is drawdown-*severity* forecasting, distinct from the terminal-death signal.

Table 5: VaR partial $\rho(\mathrm{VaR}, \mathrm{drawdown} \mid \mathrm{vol})$ under three delisting codings. Baseline and last-price coincide (mid-window delistings are rare); the increment is unchanged to within ±0.02**.**

Bed	baseline (−100%)	last-price	exclude
crypto-16	+0.12	+0.12	+0.14
crypto-17	+0.19	+0.19	+0.19
crypto-18	+0.14	+0.14	+0.14
equity-94	+0.035	+0.035	+0.035
equity-05	+0.076	+0.076	+0.076

H PRE-REGISTERED COMPOSITE AND ECONOMIC BACKTEST

The composite is $z(\mathrm{vol}) + z(\mathrm{VaR})$, equal weight, computed cross-sectionally per date, with no fitting. VaR was chosen a priori as the magnitude metric surviving correction on all five beds (§4); the weights are untuned, so the fully out-of-sample evidence is the locked-panel test (Table 4, §5), which is not significant. The economic backtest (Figure 7) sorts the universe into risk terciles and measures realized-drawdown discrimination (forward DD of the high- minus low-risk tercile) and the forward-return spread, net of a 20 bps turnover cost.

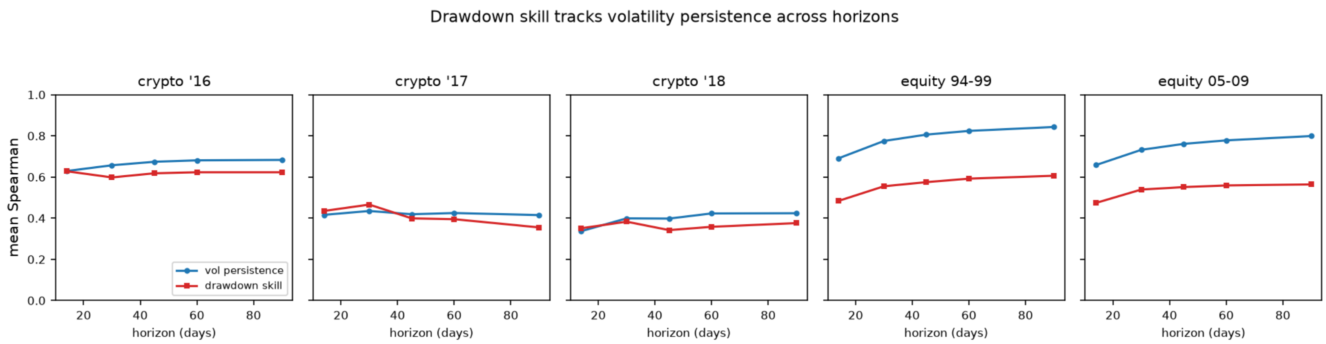

Figure 5: Volatility persistence (blue) and drawdown-forecast rank correlation (red) versus forecast horizon, by bed. The curves track.

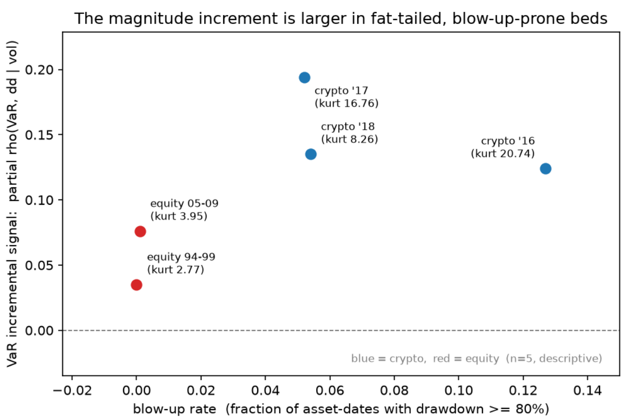

Figure 6: VaR incremental signal (partial ρ | volatility) versus blow-up rate, by bed. Behind it, a pooled per-date regression across all 144 dates gives a positive slope whose date-level CI excludes zero, suggestive when clustered by bed.

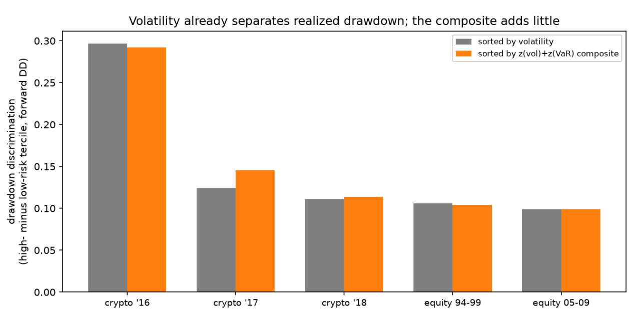

Figure 7: Realized drawdown discrimination (high- minus low-risk tercile) by bed, sorting on volatility vs. the composite. Volatility already separates drawdown strongly; the composite is indistinguishable except marginally on crypto-bull.

I DISCIPLINES ABLATION

Each of the three fair-field disciplines materially affects the reported results. We ask which shape metric one would have credited without each.

Without survivorship-free data. We rebuild the crypto universe survivors-only, keeping at each formation date only coins that neither delisted nor crashed to dust (ended below 5% of their all-time peak) by the panel end. About a third of the point-in-time top 100 survive. On the 2018 crash bed this flips negative semibeta, a co-crash shape metric, from a head-to-head -0.067 against volatility (CI $[-0.131, -0.003]$, significantly worse) to $+0.055$ (CI $[-0.082, 0.178]$, a statistical tie); downside beta closes from -0.23 to -0.01 over the same change, and VaR's partial rises from $+0.135$ to $+0.205$. Survivorship bias erases shape's penalty and inflates the magnitude edge, and it does so precisely in the crash regime, where hiding the coins that died most reshapes the cross-section. On the calm and bull crypto beds the shift is small and mixed. It does not manufacture a winner. No shape metric significantly beats volatility even survivors-only, and the rest stay far below it, the Viole–Nawrocki ratio near -1.2 and lower-tail dependence near -1.1.

Without multiple-testing correction. Across the ~45 head-to-head cells the only metrics nominally beating volatility at $p < 0.05$ are downside deviation and VaR, both tail-magnitude measures, and only on the crypto-bull bed. No shape metric is even nominally positive, so Benjamini–Hochberg is not what holds shape back.

Without a locked holdout. The discovery experiment (App. K) makes this concrete. A validation-selected metric's inflated validation score reads as a win, while its once-scored locked-test score sits at the volatility baseline.

Even stripped of every guardrail, no shape metric beats volatility.

J TARGET SENSITIVITY

Drawdown is a magnitude-dominated statistic, so one might ask whether "magnitude beats shape" is forced by the target. We re-score every metric against shape-sensitive forward targets, namely forward return skew (sign-flipped so higher is more left-asymmetric), the forward downside fraction (forward downside deviation over forward volatility, which is scale-free), and the forward tail-exceedance frequency (share of forward days below the trailing fifth percentile).

The picture inverts by target (Table 6). On the magnitude target, volatility wins on all five beds and no shape metric beats it. On every shape-sensitive target, volatility's rank correlation is near zero or negative, and the shape metrics beat it on nearly every bed. The Viole–Nawrocki ratio, the Hill exponent, both semibetas, downside beta, lower-tail dependence, and the disappointment-aversion proxy all forecast forward shape better than volatility does. The shape metrics are therefore not mis-implemented or strawmanned; they carry genuine shape information. They add nothing to drawdown forecasting because drawdown is a magnitude phenomenon and shape is not what drives it.

These are daily adaptations of metrics built at other horizons, and two deserve a note. Our per-asset Hill loss-tail exponent is a simplification of the Kelly–Jiang common tail factor, which is estimated from the pooled cross-section. We compute it per asset for a like-for-like per-date comparison. The Viole–Nawrocki upper-to-lower partial-moment ratio is high when upside dominates downside, so its negative correlation with drawdown is expected by construction, and the finding is that its *increment* beyond volatility is zero, not that its sign is wrong. That every shape metric forecasts forward shape (above) is direct evidence the adaptations are faithful rather than strawmen. Downside deviation, our operationalization of the Sortino and Bawa–Fishburn semi-deviation, is the lower partial moment of order two about a zero target.

Table 6: Winner by forward target. Volatility wins the magnitude target; shape metrics win every shape-sensitive target.

Forward target	vol rank corr (range)	shape beats vol
max drawdown (magnitude)	+0.36 to +0.62	none, any bed
−skew (shape)	−0.21 to −0.07	most, every bed
downside fraction (shape)	−0.25 to −0.05	most, every bed
tail-exceedance freq (shape)	−0.52 to −0.10	most, every bed

K AUTOMATED DISCOVERY: BEST-OF-N AND ROBUSTNESS

On the original locked panel, both the LLM loop and the 20,000-config sweep inflate the best validation score with search intensity while the locked test score stays flat at the volatility baseline (Table 7, Figure 3). Validation-selected winners beat volatility on the locked test only 43% of the time (directed search) despite a 0.94 validation–test rank correlation.

To check that this is not a single-panel artifact, we re-ran the candidate search (a general nonlinear generator over per-date feature ranks, 1,500 candidates per run) over a grid of holdout panels × 3 search seeds, in two families: *out-of-time*, a fixed-length test window slid through the timeline with validation always chronologically earlier (six windows, including the original 2017 → 2018 split); and *random*, order-ignoring 10-val/7-test repartitions of the non-train dates (six draws). Per run we record the validation-selected metric's locked-test edge over volatility (*test gap*) and the fraction of validation-winners that also beat volatility on test (*win-rate*).

Table 7: Best-of-N discovery. Validation inflates with search intensity; the locked test does not (volatility baseline 0.376). Validation and test are different splits, so compare trends, not levels.

Search intensity N	Best validation	Its locked test
1	0.27	0.31
100	0.35	0.37
20,000	0.356	0.373

The plateau is stable. The test gap averages +0.020 (out-of-time; range −0.028 to +0.046) and +0.035 (random; +0.014 to +0.072), always small and often straddling zero, so no search beats volatility out-of-sample by more than noise on any panel or seed. Validation–test rank correlation is 0.94–0.99 throughout. *The win-rate is not.* It averages 0.79 (out-of-time, range 0.55–1.00) and 0.82 (random, 0.62–0.97), lowest on holdouts where volatility's own rank correlation is high on the test (the 2018 crash, 0.43 directed / 0.57 milder generator) and near 1.0 on calm windows. Because these nominal "wins" come with a test gap of essentially zero, the *sign* of the win-rate is far less informative than its *magnitude*. The validation-selected metric matches, but does not beat, volatility.